Week2:Open Source and Dataset

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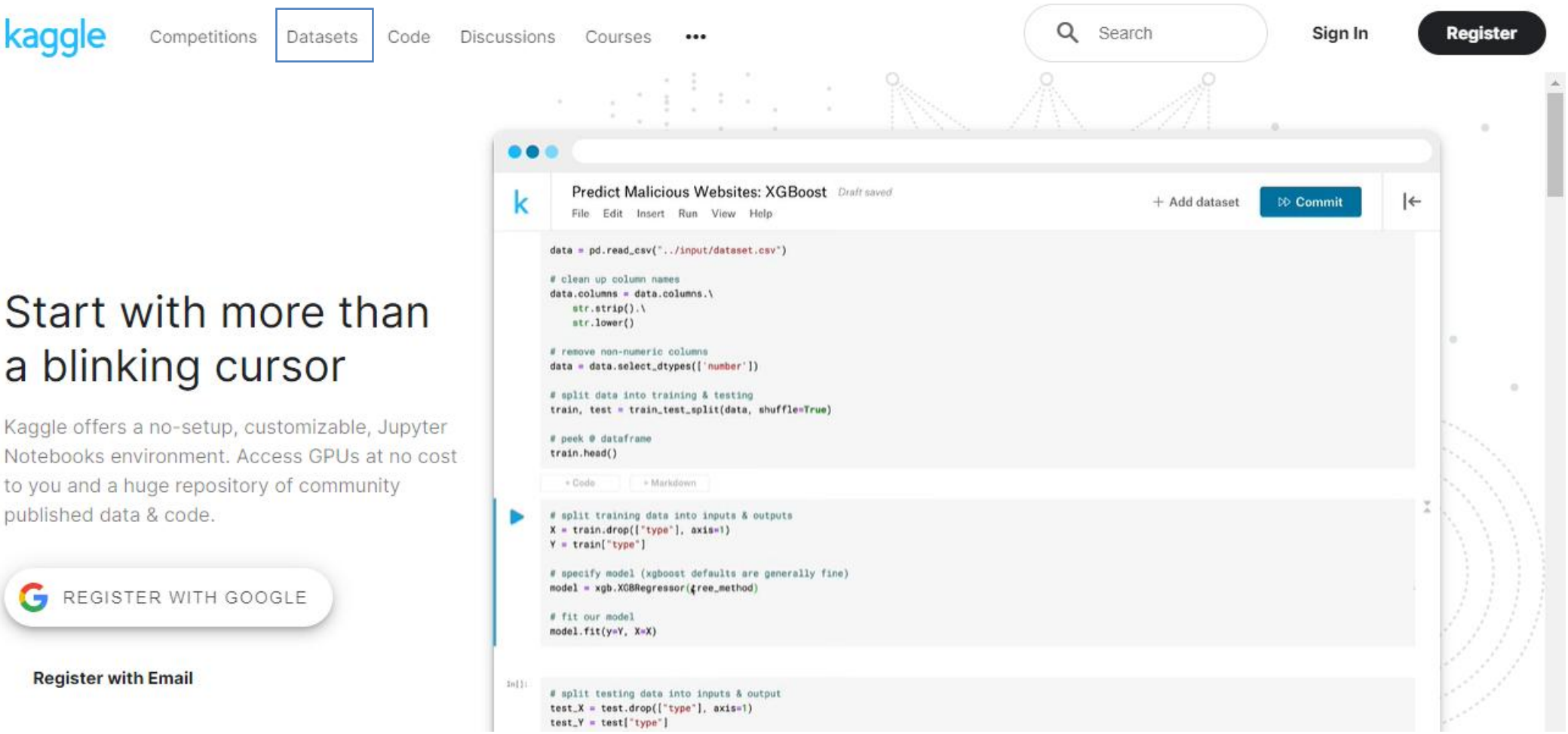
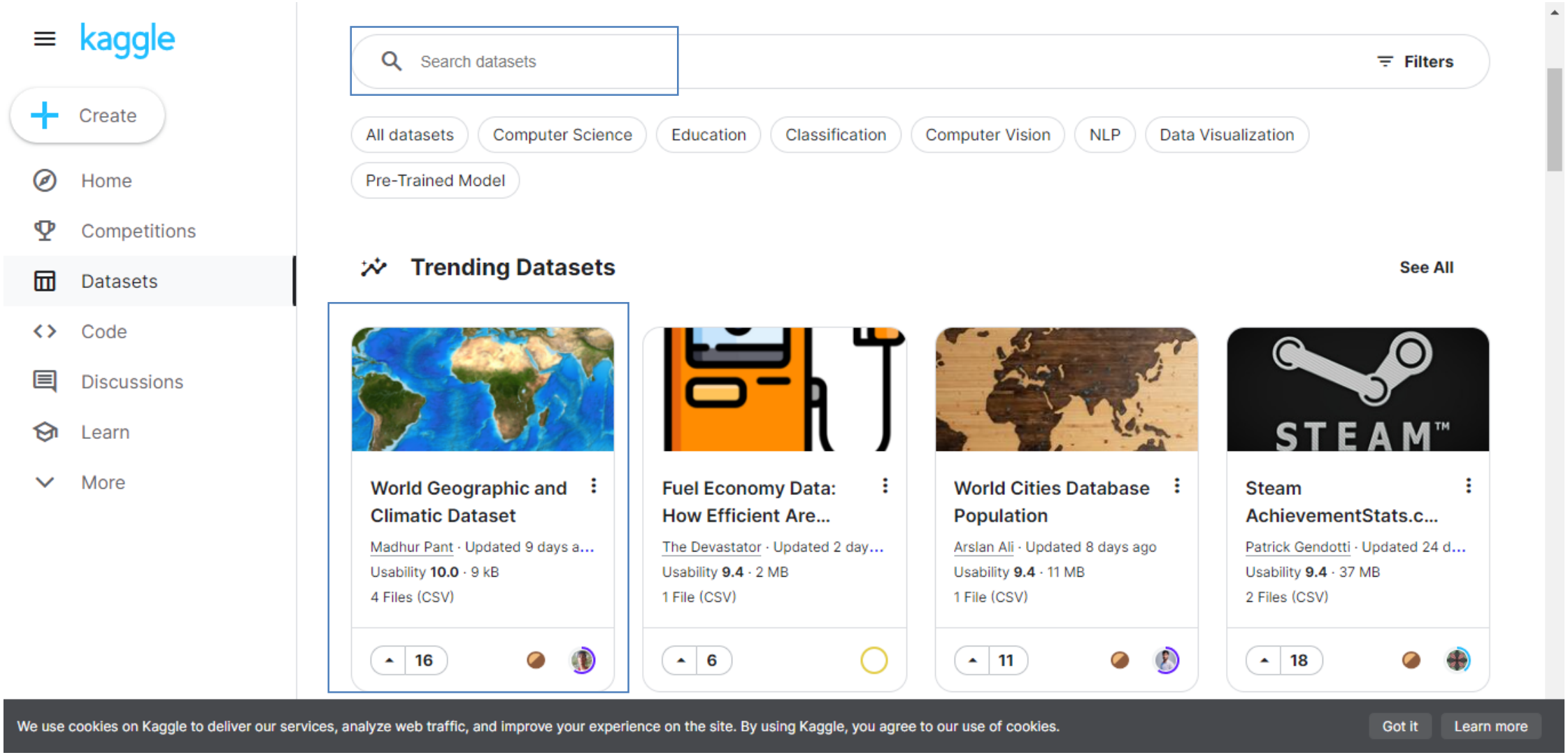


Kaggle

Choose your Databases



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Basic Syntax

How to read data?

函数	说明
read_csv	从文件、URL、文件型对象中加载带分隔符的数据。默认分隔符为逗号
read_table	从文件、URL、文件型对象中加载带分隔符的数据。默认分隔符为制表符('\t')
read_fwf	读取定宽列格式数据（也就是说，没有分隔符）
read_clipboard	读取剪贴板中的数据，可以看做 read_table 的剪贴板版。再将网页转换为表格时很有用
read_excel	从 Excel XLS 或 XLSX file 读取表格数据
read_hdf	读取 pandas 写的 HDF5 文件
read_html	读取 HTML 文档中的所有表格
read_json	读取 JSON (JavaScript Object Notation)字符串中的数据
read_msgpack	二进制格式编码的 pandas 数据
read_pickle	读取 Python pickle 格式中存储的任意对象
read_sas	读取存储于 SAS 系统自定义存储格式的 SAS 数据集
read_sql	（使用 SQLAlchemy）读取 SQL 查询结果为 pandas 的 DataFrame
read_stata	读取 Stata 文件格式的数据集
read_feather	读取 Feather 二进制文件格式

Basic Syntax

- 用于数据整理和清理、子集构造和过滤、转换等快速的矢量化数组运算
- 常用的数组算法，如排序、唯一化、集合运算等
- 高效的描述统计和数据聚合/摘要运算
- 将条件逻辑表述为数组表达式（而不是带有if-elif-else分支的循环）
- 数据的分组运算（聚合、转换、函数应用等）

Introduce to NumPy
(Numerical Python)

```
import numpy as np
```

Basic Syntax

Introduce to Pandas

```
import pandas as pd
```

- Pandas是专门为处理表格和混杂数据设计的。而NumPy更适合处理统一的数值数组数据

- Pandas经常和其它工具一同使用，如数值计算工具NumPy，分析库statsmodels和scikit-learn和数据可视化库**matplotlib和 plotly**

Data Cleaning

```
import pandas as pd
df1 = pd.DataFrame({
    'name': ['John Smith', 'Jane Doe', 'Joe Schmo'],
    'address': ['123 Main St.', '456 Maple Ave.', '789 Broadway'],
    'age': [34, 28, 51]
})
print df1
```

name	address	age
John Smith	123 Main St.	34
Jane Doe	456 Maple Ave.	28
Joe Schmo	789 Broadway	51

Data Frame

Data Frame: Row, Column, Name, String, Index, Integer

Data Cleaning

```
name,cake_flavor,frosting_flavor,topping
Devil's Food,chocolate,chocolate,chocolate shavings
Birthday Cake,vanilla,vanilla,rainbow sprinkles
Carrot Cake,carrot,cream cheese,almonds
```

name	cake_flavor	frosting_flavor	topping
Devil's Food	chocolate	chocolate	chocolate shavings
Birthday Cake	vanilla	vanilla	rainbow sprinkles
Carrot Cake	carrot	cream cheese	almonds

Introduce to ‘CSV’ File

- 纯文本，使用某个字符集，比如ASCII、Unicode、EBCDIC或GB231
- 由记录组成（典型的是每行一条记录）
- 每条记录被分隔符分隔为字段（典型分隔符有逗号、分号或制表符；有时分隔符可以包括可选的空格
- 每条记录都有同样的字段序列

Data Cleaning

Read CSV Files

```
import pandas as pd
pd.read_csv('sample.csv')
df=pd.read_csv('sample.csv')
df.to_csv('sample.csv')
print(df)
```

	Unnamed: 0	City	Population	Median Age
0	0	Maplewood	100000	40
1	1	Wayne	350000	33
2	2	Forrest Hills	300000	35
3	3	Paramus	400000	55
4	4	Hackensack	290000	39

Data Cleaning

df. head () and df.info ()

```python
import pandas as pd
df = pd.read_csv('imdb.csv')
df.to_csv('imdb.csv')
print(df.head())
print(df.info())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 220 entries, 0 to 219
Data columns (total 11 columns):
Unnamed: 0            220 non-null int64
Unnamed: 0.1          220 non-null int64
Unnamed: 0.1.1        220 non-null int64
Unnamed: 0.1.1        220 non-null int64
Unnamed: 0.1.1.1      220 non-null int64
Unnamed: 0.1.1.1.1    220 non-null int64
id                    220 non-null int64
name                  220 non-null object
genre                 220 non-null object
year                  220 non-null int64
imdb_rating           220 non-null float64
dtypes: float64(1), int64(8), object(2)
memory usage: 19.0+ KB
None
```

Data Cleaning

Select Column Data:

```
import pandas as pd

df = pd.DataFrame([
  ['January', 100, 100, 23, 100],
  ['February', 51, 45, 145, 45],
  ['March', 81, 96, 65, 96],
  ['April', 80, 80, 54, 180],
  ['May', 51, 54, 54, 154],
  ['June', 112, 109, 79, 129]],
  columns=['month', 'clinic_east',
           'clinic_north', 'clinic_south',
           'clinic_west']
)
```

```
clinic_north=df.clinic_north
```

or

```
clinic_north=df['clinic_north']
```

```
print(type(clinic_north))
```

Select Multi-Column Data:

```
clinic_north_south=df[['clinic_north','clinic_south']]
print(type(clinic_north_south))
```

Data Cleaning

Select Row Data:

```
import pandas as pd

df = pd.DataFrame([
  ['January', 100, 100, 23, 100],
  ['February', 51, 45, 145, 45],
  ['March', 81, 96, 65, 96],
  ['April', 80, 80, 54, 180],
  ['May', 51, 54, 54, 154],
  ['June', 112, 109, 79, 129]],
  columns=['month', 'clinic_east',
           'clinic_north', 'clinic_south',
           'clinic_west']
)
march=loc[2]
```

```
March_to_June=df.loc[2:5]
```

选择3月到5月的数据

```
Before_March=df.loc[:2]
print(Before_March)
```

1到2月份数据

Data Cleaning

Select Rows by Conditional Statements:

```
staff[staff.age==30]
```

选择出年龄为30岁的所有员工

```
staff[staff.age>30]
```

选择出年龄大于30岁的所有员工

```
staff[staff.age<30]
```

选择出年龄小于30岁的所有员工

```
df[df.name != 'Clara Oswald']
```

选择除员工Clara Oswald的所有行

```
variable=df[(df.age < 30) | (df.name == 'Martha Jones')]
print(variable)
```

！ = 表示不等于，I代表或，&代表并且

Data Cleaning

Lost your data ?

Data Cleaning

Lost your data ?

Albania	2016	4.511	9.437	0.638	68.100	0.730	-0.017	0.901	0.675	0.322
Albania	2017	4.640	9.476	0.638	68.400	0.750	-0.029	0.876	0.669	0.334
Albania	2018	5.004	9.518	0.684	68.700	0.824	0.009	0.899	0.713	0.319
Albania	2019	4.995	9.544	0.686	69.000	0.777	-0.099	0.914	0.681	0.274
Albania	2020	5.365	9.497	0.710	69.300	0.754	0.007	0.891	0.679	0.265
Algeria	2010	5.464	9.287		64.500	0.593	-0.205	0.618		
Algeria	2011	5.317	9.297	0.810	64.660	0.530	-0.181	0.638	0.550	0.255
Algeria	2012	5.605	9.311	0.839	64.820	0.587	-0.172	0.690	0.604	0.230
Algeria	2014	6.355	9.335	0.818	65.140				0.626	0.177
Algeria	2016	5.341	9.362	0.749	65.500				0.661	0.377
Algeria	2017	5.249	9.354	0.807	65.700	0.437	-0.167	0.700	0.642	0.289
Algeria	2018	5.043	9.348	0.799	65.900	0.583	-0.146	0.759	0.591	0.293

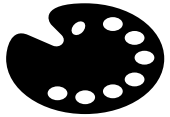
Data Cleaning

Select only the dimension you want in multi-dimensional data

127	Myanmar	Southeast	4.426	0.052	4.527	4.324	8.541	0.779	59.302	0.876	0.509	0.660	2.430	0.666	0.713	0.341	0.601	0.520
128	Jordan	Middle Eas	4.395	0.062	4.516	4.273	9.182	0.767	67.000	0.755	-0.167	0.705	2.430	0.890	0.685	0.583	0.455	0.079
129	Chad	Sub-Sahar	4.355	0.094	4.540	4.171	7.364	0.619	48.478	0.579	0.041	0.807	2.430	0.255	0.353	0.000	0.240	0.215
130	Sri Lanka	South Asia	4.325	0.066	4.454	4.196	9.470	0.827	67.299	0.841	0.079	0.863	2.430	0.990	0.820	0.593	0.559	0.239
131	Swaziland	Sub-Sahar	4.308	0.071	4.448	4.168	9.065	0.770	50.833	0.647	-0.185	0.708	2.430	0.849	0.693	0.074	0.323	0.067
132	Comoros	Sub-Sahar	4.289	0.084	4.454	4.123	8.031	0.626	57.349	0.548	0.082	0.781	2.430	0.488	0.367	0.279	0.202	0.241
133	Egypt	Middle Eas	4.283	0.045	4.371	4.195	9.367	0.750	61.998	0.749	-0.182	0.795	2.430	0.954	0.647	0.426	0.446	0.069
134	Ethiopia	Sub-Sahar	4.275	0.051	4.374	4.175	7.694	0.764	59.000	0.752	0.082	0.761	2.430	0.370	0.679	0.331	0.451	0.241
135	Mauritania	Sub-Sahar	4.227	0.070	4.365	4.090	8.542	0.795	57.161	0.561	-0.106	0.731	2.430	0.666	0.749	0.273	0.218	0.119
136	Madagascar	Sub-Sahar	4.208	0.072	4.349	4.068	7.396	0.686	59.305	0.552	-0.005	0.803	2.430	0.266	0.503	0.341	0.207	0.185
137	Togo	Sub-Sahar	4.107	0.077	4.258	3.956	7.362	0.569	54.914	0.619	0.032	0.772	2.430	0.254	0.239	0.203	0.289	0.209
138	Zambia	Sub-Sahar	4.073	0.069	4.209	3.938	8.145	0.708	55.809	0.782	0.061	0.823	2.430	0.528	0.552	0.231	0.487	0.227
139	Sierra Leone	Sub-Sahar	3.849	0.077	4.001	3.698	7.434	0.630	51.651	0.717	0.084	0.866	2.430	0.279	0.377	0.100	0.408	0.243
140	India	South Asia	3.819	0.026	3.869	3.769	8.755	0.603	60.633	0.893	0.089	0.774	2.430	0.741	0.316	0.383	0.622	0.246
141	Burundi	Sub-Sahar	3.775	0.107	3.985	3.565	6.635	0.490	53.400	0.626	-0.024	0.607	2.430	0.000	0.062	0.155	0.298	0.172
142	Yemen	Middle Eas	3.658	0.070	3.794	3.521	7.578	0.832	57.122	0.602	-0.147	0.800	2.430	0.329	0.831	0.272	0.268	0.092
143	Tanzania	Sub-Sahar	3.623	0.071	3.762	3.485	7.876	0.702	57.999	0.833	0.183	0.577	2.430	0.433	0.540	0.300	0.549	0.307
144	Haiti	Latin Amer	3.615	0.173	3.953	3.276	7.477	0.540	55.700	0.593	0.422	0.721	2.430	0.294	0.173	0.227	0.257	0.463
145	Malawi	Sub-Sahar	3.600	0.092	3.781	3.419	6.958	0.537	57.948	0.780	0.038	0.729	2.430	0.113	0.168	0.298	0.484	0.213
146	Lesotho	Sub-Sahar	3.512	0.120	3.748	3.276	7.926	0.787	48.700	0.715	-0.131	0.915	2.430	0.451	0.731	0.007	0.405	0.103
147	Botswana	Sub-Sahar	3.467	0.074	3.611	3.322	9.782	0.784	59.269	0.824	-0.246	0.801	2.430	1.099	0.724	0.340	0.539	0.027
148	Rwanda	Sub-Sahar	3.415	0.068	3.548	3.282	7.676	0.552	61.400	0.897	0.061	0.167	2.430	0.364	0.202	0.407	0.627	0.227
149	Zimbabwe	Sub-Sahar	3.145	0.058	3.259	3.030	7.943	0.750	56.201	0.677	-0.047	0.821	2.430	0.457	0.649	0.243	0.359	0.157
150	Afghanistan	South Asia	2.523	0.038	2.596	2.449	7.695	0.463	52.493	0.382	-0.102	0.924	2.430	0.370	0.000	0.126	0.000	0.122

Tips:

Sketch the charts in your mind.



Pay great attention to color.

Be brave and innovative.

Thanks for listening!